MIL-C-60604 (MU)
16 February 1967
SUPERSEDING
FA-PD-MI-2328, Rev. 3
31 October 1962

# MILITARY SPECIFICATION
# CARTRIDGE, THRUSTER, M119

## 1. SCOPE

1.1 This specification covers Cartridge, Thruster, M119 for use with propellant actuated devices in aircrew emergency escape systems.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS
Military

| | |
|---|---|
| MIL-A-2550 | - Ammunition and Special Weapons, General Specifications for |
| MIL-E-5272 | - Environmental Testing, Aeronautical and Associated Equipment, General Specification for |
| MIL-H-25579 | - Hose Assembly, Tetrafluoroethylene, High Temperature, Power Plant, Aircraft |
| MIL-C-60539 | - Coating, Anodic, Conventional for Aluminum and Aluminum Alloys |

STANDARDS
Military

| | |
|---|---|
| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-109 | - Quality Assurance Terms and Definitions |
| MIL-STD-639 | - Surface Roughness Values Preferred for Army Ordnance Materiel |
| MIL-STD-1167 | - Ammunition Data Card |
| MS-28741 | - Hose Assembly, Detachable End Fitting, Medium Pressure |

FSC 1377

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DRAWINGS
Munitions Command

| | |
|---|---|
| D8595513 | - Cartridge, Thruster M119, Assembly |
| C8830980 | - Carton, Packing, Ammunition for Cartridge, Thruster, M119 |
| C8830981 | - Box, Packing, Ammunition for Cartridge, Thruster, M119 |
| IEL-8595513 | - Index of Inspection Equipment List for Cartridge, Thruster, M119 |

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement function shall be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall conform to the applicable drawings, referenced specifications and the following:

3.2 First article sample.- Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.3 Protective finish.- Protective finish shall be in accordance with MIL-C-60539.

3.4 Vibration.- The assembled cartridge shall withstand vibration as prescribed in MIL-E-5272, Procedure XII, without evidence of any disintegration of the primer, propellant or igniter, or derangement of any component.

3.5 Functional

3.5.1 The cartridge shall not fail to fire.

3.5.2 Ballistics.- The cartridge when tested in either the M9, M13 or M19 thruster, temperature conditioned at -65° Fahrenheit (F) to -70°F and at 200°F to 205°F shall comply with the following requirements:

a. Thruster M9 - Shall propel a 50 pound mass horizontally against a resistive force of 800 pounds, minimum for a stroke of 3.6 inches, minimum to 3.8 inches maximum. Time to complete thruster stroke shall not exceed 80 milliseconds. The thruster shall produce a bypass pressure of 1000 pounds per square inch (PSI) minimum, at the end of 42 inches of aircraft hose, size 4 Standard MS28741 when the stroke has reached a minimum of 3.6 inches. Ignition delay time shall not exceed 25 milliseconds (See 6.1). The thruster shall be fired by the gas pressure generated from an M3A1 initiator connected to the thruster with 15 feet of aircraft hose, size 4 in accordance with MS-28741.



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b. Thruster M13 - With the thruster connected to a 70 pound weight, the piston shall push the weight vertically and, in addition, shear a 2000 pound shear pin during the first movement of the weight. The thrust developed shall not exceed 5000 pounds. The thruster shall produce a by-pass pressure of 1000 pounds per square inch minimum at the end of 6 feet of aircraft hose, size 4 in accordance with MIL-H-25579 and shall complete a piston stroke ranging from 2.76 inches minimum to 2.86 inches maximum. Time to complete thruster stroke shall not exceed 80 milliseconds. Ignition delay time shall not exceed 25 milliseconds (See 6.1). The thruster shall be fired by the gas pressure generated from an M3A1 initiator connected to the thruster with 15 feet of aircraft hose, size 4 in accordance with MS-28741.

c. Thruster M19 - With the thruster connected to a 75 pound weight, the piston shall push the weight along a horizontal path of travel and the moving piston and weight shall overcome a resisting force of from 25 pounds at zero stroke to 500 pounds at 2.40 inches of stroke. The thrust developed shall not exceed 2500 pounds after an elapsed time of 16 milliseconds. The thruster shall produce a by-pass pressure of 1000 pounds per square inch (PSI) minimum, at the end of a 13 foot 6 inch length of aircraft hose, size 4 in accordance with MIL-H-25579 and shall complete a piston stroke ranging from 2.76 inches, minimum to 2.86 inches, maximum. Time to complete thruster stroke shall be 48 milliseconds, minimum, and 80 milliseconds maximum. Ignition delay time shall not exceed 25 milliseconds (see 6.1). The thruster shall be fired by the gas pressure generated from an M3A1 initiator connected to the thruster with 15 feet of aircraft hose, size 4 in accordance with MS-28741.

3.6 Workmanship.- Workmanship shall be in accordance with the requirements of the applicable drawings, referenced specifications and the following:

3.6.1 Metal defects.- The metal parts shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.6.2 Burr.- The part or assembly shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.6.3 Foreign matter.- The part or assembly shall not contain chips, dirt, grease, oil, rust, corrosion or other foreign matter.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality Assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550.

4.2 First article sample.

4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 35 assembled cartridges, 50 unplated cartridge cases for hardness tests, 10 sets of unassembled parts (including explosive components) for visual and dimensional examination and 23 sets of unassembled components without explosive components for protective finish tests.

TABLE II

| Paragraph Requirement | Test | Test | Sample Size | Type of Test Destr | Nondestr |
|---|---|---|---|---|---|
| 3.3 | 4.4.1 | Protective Finish | 23 | X | - |
| 3.4 | 4.4.2 | Vibration | 12 (fr Ballistics) | - | X |
| Dwg. | 4.4.3 | Hardness | 50 (Unplated) | X | - |
| Dwg. | 4.4.4 | Airtightness | Each | - | X |
| Dwg. | 4.4.5 | Sealant application | 5 | X | - |
| 3.5.2 | 4.4.6.1 | Ballistics | 30 | X | - |

The sample shall be manufactured using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply.



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4.2.1.1 Examination and tests.- After acceptance and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a government laboratory or such other facility specified in the contract.

4.2.1.2 Ballistics.- The sample for ballistic tests shall be temperature conditioned and tested to determine compliance with 3.5.2. The sample for firing shall consist of the following:

TABLE III

| Temperature | Sample | Vibrated |
|---|---|---|
| -65°F to -70°F | 6 | 4 |
| 65°F to 75°F | 6 | 4 |
| 200°F to 205°F | 6 | 4 |

4.2.1.3 Initial production sample failure.- Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Size of lot.- Unless otherwise specified by the procuring agency, a lot shall consist of 1200 cartridges, maximum, exclusive of test samples and shall consist of:

a. Components produced by a homogeneous process

b. Propellant from not more than 1 lot

c. Primers from not more than 1 lot

d. Igniter from not more than 1 lot



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4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Tables IV and V. For an individual defect (single characteristics) the criteria in Table IV shall be used. For total defects within a major or minor class the criteria in Table V shall be used. All non-conforming material shall be rejected. Inspection for rubber O-rings shall be in accordance with MIL-STD-413. The Ordnance Standard Finish Blocks prescribed in MIL-STD-639 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

TABLE IV

| Lot Size | Sample Size | Major | | Minor | |
|---|---|---|---|---|---|
| | | Accept | Reject | Accept | Reject |
| 1 - 300 | 100% examination | - | - | - | - |
| 301 - 1300 | 110 | 1 | 2 | 2 | 3 |
| 1301 - 3200 | 150 | 2 | 3 | 3 | 4 |
| 3201 - 8000 | 225 | 3 | 4 | 4 | 5 |
| 8001 - 22000 | 300 | 4 | 5 | 5 | 6 |

TABLE V

| Sample Size for Major Defects Number of Defects in class | Total Allowable Defects Per Sample Size | | | |
|---|---|---|---|---|
| | 110 | 150 | 225 | 300 |
| 2 - 3 | 1 | 2 | 3 | 4 |
| 4 - 7 | 2 | 3 | 4 | 5 |
| 8 - 19 | 3 | 4 | 5 | 7 |
| 20 - 43 | 4 | 5 | 8 | 10 |
| Over 43 | 6 | 8 | 11 | 14 |
| **Sample Size for Minor Defects Number of Defects in class** | 110 | 150 | 225 | 200 |
| 2 - 3 | 2 | 3 | 4 | 5 |
| 4 - 7 | 3 | 4 | 5 | 7 |
| 8 - 16 | 4 | 5 | 8 | 10 |
| 17 - 48 | 6 | 8 | 11 | 14 |
| Over 48 | 8 | 11 | 17 | 20 |



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4.3.2.1 Primer, 72M, C7645332, covering a detail of Dwg D8595513

| Defect | Method of Inspection |
|---|---|
| **CRITICAL** | |
| 1. Mixed primers | Visual |
| 2. Primer pellet dislodged | Visual |
| 3. Primer foil discolored | Visual |
| 4. Metal defective | Visual |
| **MAJOR** | |
| 101. Burr | Visual |
| 102. Foreign matter | Visual |
| **MINOR** | |
| None defined | |

4.3.2.2 Case, Cartridge, Dwg C8595514, covering a detail of Dwg D8595513

| | |
|---|---|
| **CRITICAL** | |
| None defined | |
| **MAJOR** | |
| 101. Concentricity of head with body | Gage |
| 102. Straightness of base | Gage |
| 103. Diameter of body, min. | Gage |
| 104. Diameter of head, min. | Gage |
| 105. Thickness of wall | Gage |
| 106. Thickness of base flat | Gage |
| 107. Location of datum diameter | Gage |
| 108. Metal defective | Visual |
| 109. Protective finish inadequate or defective | Visual |
| 110. Burr | Visual |
| 111. Foreign matter | Visual |



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4.3.2.2 (Cont'd)

| Defect | Method of Inspection |
|---|---|
| **MINOR** | |
| 201. Diameter of body, max. | Gage |
| 202. Diameter of head, max. | Gage |
| 203. Total length | Gage |
| 204. Radius or chamfer missing or incorrect | Visual |
| 205. Surface finish improper | Visual |

4.3.2.3 Cartridge Head, Dwg C8595515, covering a detail of Dwg D8595513

CRITICAL

None defined

| | |
|---|---|
| **MAJOR** | |
| 101. Concentricity of flange with head | Gage |
| 102. Concentricity of small center hole with primer pocket | Gage |
| 103. Concentricity at chamfer mouth with body | Gage |
| 104. Concentricity of body with head | Gage |
| 105. Concentricity of head with primer pocket | Gage |
| 106. Diameter of small center hole, max. | Gage |
| 107. Diameter of primer pocket | Gage |
| 108. Depth of primer pocket | Gage |
| 109. Thickness of head at primer pocket | Gage |
| 110. Location of flange | Gage |
| 111. Location of datum diameter | Gage |
| 112. Metal defective | Visual |
| 113. Protective finish inadequate or defective | Visual |
| 114. Surface finish of primer pocket | Visual |
| 115. Burr or sharp corner except where permitted | Visual |
| 116. Foreign matter | Visual |
| **MINOR** | |
| 201. Diameter of flange | Gage |
| 202. Diameter of small center hole, min. | Gage |
| 203. Diameter of body | Gage |



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4.3.2.3 (Cont'd)

| Defect | Method of Inspection |
|---|---|
| MINOR (Cont'd) | |
| 204. Diameter at chamfer mouth | Gage |
| 205. Diameter of head | Gage |
| 206. Depth to base of primer pocket | Gage |
| 207. Total length | Gage |
| 208. Radius or chamfer missing or incorrect | Visual |
| 209. Surface finish improper, except as otherwise classified | Visual |

4.3.2.4 Ring, Cartridge B8595516, covering a detail of Dwg D8595513

CRITICAL

None defined

| | |
|---|---|
| MAJOR | |
| 101. Concentricity of inside with outside diameter | Gage |
| 102. Diameter outside, min. | Gage |
| 103. Metal defective | Visual |
| 104. Protective finish | Visual |
| 105. Burr | Visual |
| 106. Foreign matter | Visual |
| MINOR | |
| 201. Diameter inside | Gage |
| 202. Diameter outside, max. | Gage |
| 203. Total length | Gage |
| 204. Radius missing or incorrect | Visual |
| 205. Surface finish, improper | Visual |

4.3.2.5 Gasket, Dwg B8595517, covering a detail of Dwg D8595513

CRITICAL

None defined



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4.3.2.5 (Cont'd)

| Defect | Method of Inspection |
|---|---|
| MAJOR | |
| 101. Diameter, min. | Gage |
| 102. Material defective | Visual |
| 103. Flash | Visual |
| 104. Foreign matter | Visual |
| MINOR | |
| 201. Diameter, max. | Gage |
| 202. Width | Gage |
| 203. Thickness | Gage |

4.3.2.6 Cartridge, Thruster, M119 Assembly, Dwg D8595513

| | |
|---|---|
| CRITICAL | |
| 1. Primer not below flush with surface "S" | Gage |
| 2. Primer not fully seated to bottom of primer pocket | X-Ray Visual 1/ |
| 3. Any part missing, except as otherwise classified | X-Ray Visual 1/ |
| 4. Improper assembly | X-Ray Visual 1/ |
| 5. Primer pellet missing | X-Ray Visual 1/ |
| 6. Anvil missing, cocked, broken or inverted | X-Ray Visual 1/ |
| 7. Marking missing, incorrect or illegible | Visual |
| MAJOR | |
| 101. Concentricity of head with body | Gage |
| 102. Diameter of head, max. | Gage |
| 103. Diameter of body, max. | Gage |
| 104. Depth to surface "S" | Gage |
| 105. Total length, max. | Gage |
| 106. Location of datum diameter | Gage |
| 107. Top edge of primer and pocket not properly sealed and crimped | Visual |
| MINOR | |
| 201. Varnish missing or inadequate | Visual |



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1/ Prior to assembly into a cartridge, the cartridge shall be x-rayed with the sides of the case perpendicular to the rays. The x-ray negatives, properly identified, shall be forwarded to the procuring activity.

4.3.2.7 Carton, Packing, Ammunition for Cartridge Thruster, M119, Dwg C8830980.

| Defect | Method of Inspection |
|---|---|
| CRITICAL | |
| None defined | |
| MAJOR | |
| None defined | |
| MINOR | |
| 201. Contents loose | Manual & Visual |
| 202. Quantity of contents incorrect | Visual |
| 203. Separator or filler missing | Visual |
| 204. Heat sealer incomplete, loose or distorted | Visual |
| 205. Cut or damage through impregnated box | Visual |
| 206. Marking missing, incorrect or illegible | Visual |

4.3.2.8 Box, Packing, Ammunition for Cartridge, Thruster, M141, Dwg C8830981.

| | |
|---|---|
| CRITICAL | |
| None defined | |
| MAJOR | |
| None defined | |
| MINOR | |
| 201. Contents loose | Manual & Visual |
| 202. Hardware missing, loose, broken or improperly engaged | Visual |
| 203. Strapping missing, loose, broken or improperly assembled | Visual |
| 204. Contents of box exposed | Visual |
| 205. Box damaged | Visual |



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4.3.2.8 (Cont'd)

| Defect | Method of Inspection |
|---|---|
| MINOR (Cont'd) | |
| 206. Marking missing, incorrect or illegible | Visual |
| 207. Number of carton incorrect | Visual |
| 208. Car seal missing, unsealed or improperly positioned | Visual |

4.3.3 Inspection testing.- The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Protective finish.- Twenty-three units of each component shall be selected for protective finish tests. Failure of any component to comply with the requirements of 3.3 shall be cause for rejection of the lot.

4.3.3.2 Vibration.- Twelve of the thirty cartridges of the initial production sample shall be subjected to vibration. The twelve cartridges shall then be x-rayed. Failure of any cartridge to comply with the requirements of 3.4 shall be cause for rejection of the lot. Any change in the process of a lot of primers in subsequent production shall be reason to select an additional sample of 12 cartridges for vibration tests.

4.3.3.3 Hardness.- Fifty units from each heat treated lot of components shall be tested. Failure of any unit to comply with the drawing requirements shall be cause for rejection of the lot of components.

4.3.3.4 Airtightness.- Each cartridge shall be subjected to the specified pressure differential. Failure to comply with the drawing requirements shall be cause for rejection of the cartridge.

4.3.3.5 Sealant application.- Five cartridges from each production lot shall be selected for this test. Failure of any cartridge to comply with the drawing requirements shall be cause for rejection of the lot of cartridges.

4.3.3.6 Functional.

4.3.3.6.1 Ballistics.- Thirty units shall be tested to determine compliance with the requirements of 3.5.2. The units shall be temperature conditioned as provided in Table VI and 4.4.6.1.1. Failure of any unit to comply with the requirements of 3.5.2 shall be cause for rejection of the lot.

TABLE VI

| Temperature | Sample Size |
|---|---|
| -65° to -70°F | 10 |
| 65° to 75°F | 10 |
| 200° to 205°F | 10 |



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4.3.4 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL-8595513.

4.4 Test methods and procedures.

4.4.1 Protective finish.- Test procedures for protective finish shall be in accordance with MIL-C-60539. The cartridge case and cartridge head shall be sectioned longitudinally to better expose the significant surfaces to the salt spray.

4.4.2 Vibration.- The method and procedure for performing the vibration test shall be in accordance with MIL-E-5272, Procedure XII.

4.4.3 Hardness.- A hardness testing machine shall be used to determine the hardness values specified on the applicable drawing.

4.4.4 Airtightness.- The method for determining airtightness shall be by use of equipment listed on IEL-8595513.

4.4.5 Sealant application.- The cartridge shall be disassembled in such a manner that determination can be made that the sealant has been applied for the full length of the joining surfaces at the junction of the cartridge head and case sidewall and that the propellant cavity is free of sealant.

4.4.6 Functional.

4.4.6.1 Ballistics.- The cartridge shall be loaded into an acceptable M9, M13 or M19 thruster and divided into 3 equal groups and temperature conditioned in accordance with Table VI. The cartridge shall be tested within ten minutes after removal from the conditioning chamber. When this time is exceeded, the thruster shall be reconditioned in accordance with 4.4.6.1.1.

4.4.6.1.1 Temperature conditioning.- A standard M9, M13 or M19 thruster loaded with a dummy M119 cartridge containing a temperature sensitive element shall be placed in the conditioning chamber at the same time as the sample. The conditioning time shall be a minimum of 1 hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.- Packing and marking shall be in accordance with drawings C8830980 and C8830981.

5.2 Data card.- Data card information shall be in accordance with MIL-STD-1167.

6. NOTES



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6.1 Ignition delay is defined as the time that elapses between the firing of the primer and the beginning of a continuous pressure rise on a pressure time record.

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# SPECIFICATION ANALYSIS SHEET

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**INSTRUCTIONS**

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

| SPECIFICATION | | |
|---|---|---|
| ORGANIZATION | CITY AND STATE | |
| CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT<br>$ |

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

| SUBMITTED BY *(Printed or typed name and activity)* | DATE |
|---|---|
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